

Hotel Management System

UI Redesign & Feature Enhancement Proposal

This document outlines the complete UI overhaul plan for HMS — transforming the current functional-but-plain interface into a modern, dark-themed, role-aware POS system designed specifically for Indian hotel and restaurant staff with minimal technical training.

8	7	11	1
UI Files to Change	Screens Redesigned	AI Agents Shown	New Agent Screen

Current State vs Target State

Current UI (Screenshots shared)	Target UI (Designed prototype)
Plain white/light grey background	Dark theme (#0E1117) with glowing accents
Minimal visual hierarchy	Cards, badges, color-coded status indicators
Simple list views (no icons, no color)	Grid menu with stock status, category tabs
Kitchen Queue: plain list, no timers	Kitchen Display: ticket cards with countdown timers
Reports: empty boxes, no charts	Reports: bar charts, payment donut, top-5 rankings
Chat screen: basic text box	AI Agent: 3-panel with EventBus log + pipeline trace
No role-based UI differences	Role-aware: discount/void hidden for waiters
No offline status indicator	Always-visible offline/online badge

The 7 Redesigned Screens

■ Login / Onboarding

PIN keypad, role selector (Waiter/Manager/Cashier/Clerk/Admin/Kitchen), brand mark with glow effect. Dark background with radial gradients. Animated PIN dots that fill on press.

- PIN-based auth with masked display
- Role chip selector (6 roles)
- Offline badge always visible
- No changes to API — same POST /api/auth/login

■ Dashboard / Home

Command center showing live KPIs, quick action buttons, active orders list, payment breakdown, and real-time audit activity feed. Replaces the empty post-login screen.

- 4 stat cards (Revenue, Orders, Low Stock, Avg Order)
- Quick action grid (New Order, Add Stock, Reports, Kitchen)
- Active orders with status badges (PENDING/COOKING/READY/LATE)
- Payment breakdown progress bars

■ POS / Order Entry

Split layout: left panel = menu grid with category tabs + search, right panel = order cart with live totals. Large visual menu cards replace the plain list. Keyboard shortcuts shown.

- Menu grid (3 cols) with stock indicators
- Category tab filter bar
- Voice mic button (existing feature)
- Same API endpoints, new visual layer only

■ Inventory / Stock

Two-column layout: alert sidebar + main stock table. Replaces the plain Menu Management list. Visual stock bars, color-coded status, and append-only ledger history section.

- Stock progress bars (green/yellow/red)
- Alert sidebar for critical/low items
- Append-only ledger table
- Category filter sidebar
- Same GET/POST `/api/inventory/*` endpoints

■ Invoice / Receipt

Side-by-side layout: left = payment method + order history table, right = formatted receipt preview. Replaces the basic receipt screen with a professional printable format.

- 3 payment method cards (Cash/Card/Voucher)
- Change calculator
- Formatted receipt with GST breakdown
- Order history table with reprint buttons
- Same POST `/api/sales/orders/{id}/finalize`

■ Reports / Analytics

Full analytics dashboard with hourly bar chart, payment donut chart, top-5 selling items, and inventory snapshot. Replaces the empty reports screen with real visualizations.

- Hourly revenue bar chart
- Payment split donut chart
- Top 5 items with progress bars
- Inventory 2x2 grid (In Stock/Low/Out/Total)
- Date picker + CSV export

■ Kitchen Display System

Full dark KDS screen optimized for kitchen staff. Ticket-card grid with color-coded urgency, countdown timers, item checkoff, and bump buttons. Replaces the plain Kitchen Queue list.

- Ticket cards: green (< 5min) / yellow (5-15min) / red (15+ min blinking)
- Countdown timers per ticket
- Mark items done individually
- Mark Ready / Bump / Served buttons
- Bottom stats bar

■ NEW: AI Agent Interaction Screen

The most impactful addition — a dedicated screen where staff can control the entire system through natural language chat or voice commands. Built on top of the existing OrchestratorAgent, InsightAgent, and IntentParser already in the codebase.

Panel	Contents	Backed by
Left Sidebar	All 11 agents with live status dots, LLM provider info, clickable example commands	AgentRegistry
Center Chat	3 modes: Ask/Insights, Command, Voice. Shows pipeline trace (STT→Intent→Orchestrator→Agents)	ChatScreen + OrchestratorAgent
Right EventBus	Live stream of all events: workflow.multi_step, order.created, payment.completed, audit.logged, etc.	EventBus log

Example Commands the Agent Understands:

- "2 biryani and 1 coke for table 5, pay cash" → Creates order + finalizes in one step
- "What are today's total sales?" → InsightAgent queries DB, LLM formats answer
- "How much paneer is left?" → InventoryAgent returns stock count
- "Finalize order for table 3 by card" → OrchestratorAgent fires finalize workflow
- "Add 50 units of rice to stock" → InventoryAgent records stock-in
- "Void order on table 2, customer cancelled" → Requires manager approval

Implementation Plan

Phase 1 — Design System & Colors

1-2 days

Update HMSColors in ui_helpers.py. Add dark theme CSS variables. Switch ft.ThemeMode to DARK. No logic changes.

Files: src/ui/components/ui_helpers.py – HMSColors class, src/ui/app.py – theme_mode, background

Phase 2 — Login & Navigation

1 day

Redesign auth_screen.py with branded logo, role chips, glowing PIN dots. Update sidenav in app.py.

Files: src/ui/screens/auth_screen.py, src/ui/app.py – NavigationRail styling

Phase 3 — Dashboard Screen

2 days

New dashboard_screen.py. Shows KPIs, quick actions, active orders, activity feed. Hook into existing APIs.

Files: src/ui/screens/dashboard_screen.py (NEW), src/ui/app.py – add dashboard route

Phase 4 — POS Screen

2 days

Refactor pos_screen.py. Replace ItemPickerWidget list with card grid. Add category tab filter. Keep all API calls.

Files: src/ui/screens/pos_screen.py, src/ui/components/ui_helpers.py – ItemPickerWidget

Phase 5 — Inventory Screen

1-2 days

Refactor products_screen.py. Add alert sidebar, stock bars, ledger history table.

Files: src/ui/screens/products_screen.py

Phase 6 — Reports Screen

2 days

Refactor reports_screen.py. Add bar chart (canvas drawing), donut chart, top-5 items ranking.

Files: src/ui/screens/reports_screen.py

Phase 7 — Kitchen Display

1 day

Refactor kitchen_screen.py. Replace list with ticket-card grid. Add timers and bump buttons.

Files: src/ui/screens/kitchen_screen.py (or order_history_screen.py)

Phase 8 — Agent Screen

2-3 days

Enhance chat_screen.py with 3-panel layout: agent sidebar, pipeline trace view, EventBus live log.

Files: src/ui/screens/chat_screen.py, src/agents/ – add event log streaming endpoint

Phase 9 — Invoice/Receipt

1 day

Refactor receipt_screen.py. Add order history table, payment method selector, formatted preview.

Files: `src/ui/screens/receipt_screen.py`, `src/ui/screens/order_history_screen.py`

Key Principle: UI-Only Changes

The redesign is intentionally **UI-layer only**. All existing FastAPI endpoints, domain logic, agent workflows, database schema, and business rules remain **100% unchanged**. Only the Flet screen files and `ui_helpers.py` are modified.

UNCHANGED (Zero risk)	CHANGED (UI layer only)
<code>src/api/app.py</code> — all 25+ endpoints	<code>src/ui/screens/*.py</code> — visual redesign
<code>src/domain/</code> — business rules, entities	<code>src/ui/components/ui_helpers.py</code> — colors, buttons
<code>src/application/services.py</code>	<code>src/ui/app.py</code> — theme + new dashboard route
<code>src/infrastructure/</code> — DB, repos	<code>src/ui/screens/chat_screen.py</code> — layout upgrade
<code>src/agents/</code> — all 11 agents	<code>src/ui/screens/dashboard_screen.py</code> (NEW)
<code>src/voice/</code> — STT, TTS, IntentParser	CSS variables + dark color scheme
All 166 tests — no changes needed	